Complex Algorithm Simulator

Jack Brunswik

February 25, 2026

Project Board Screenshots

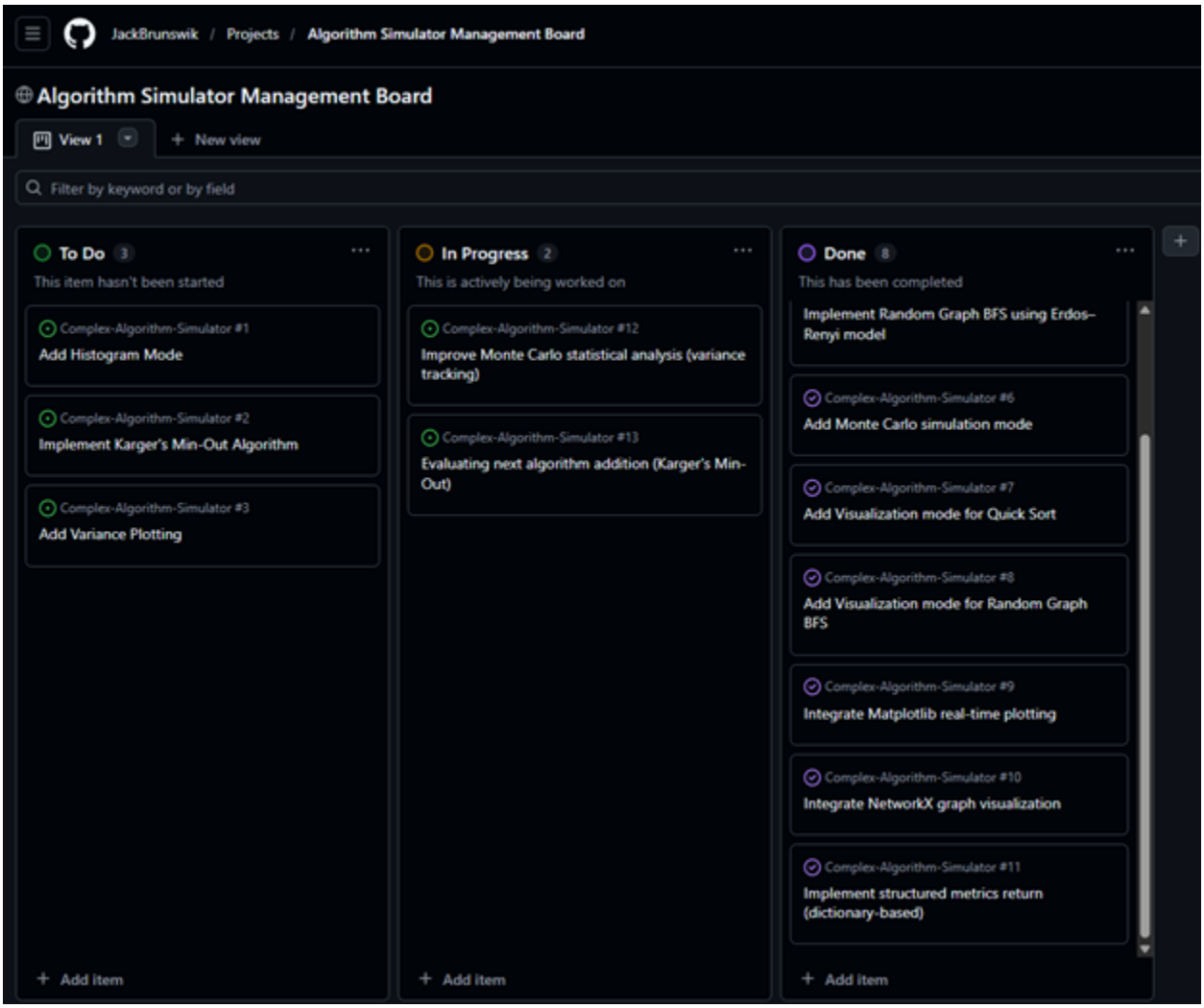

Figure 1: Overview of project board

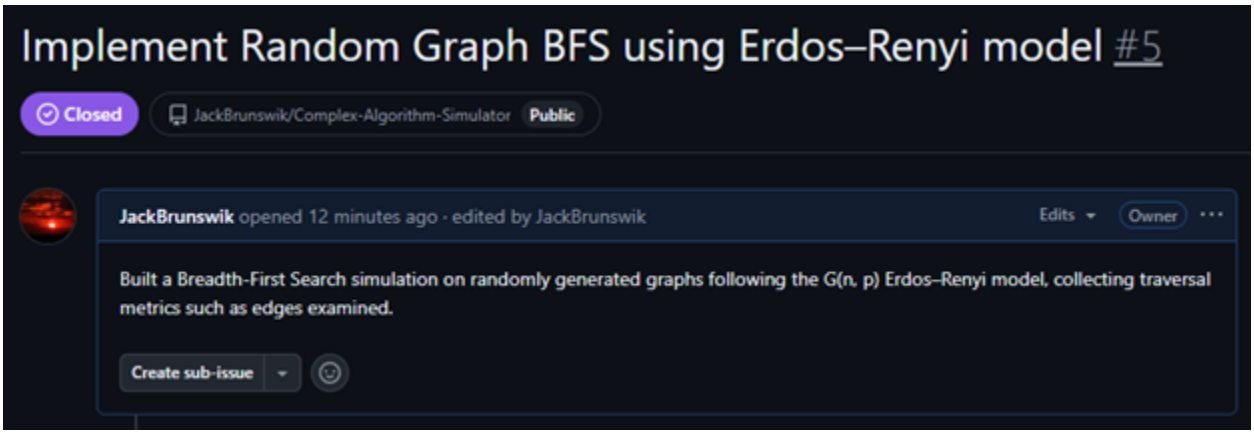

Figure 2: Brief description of completed work

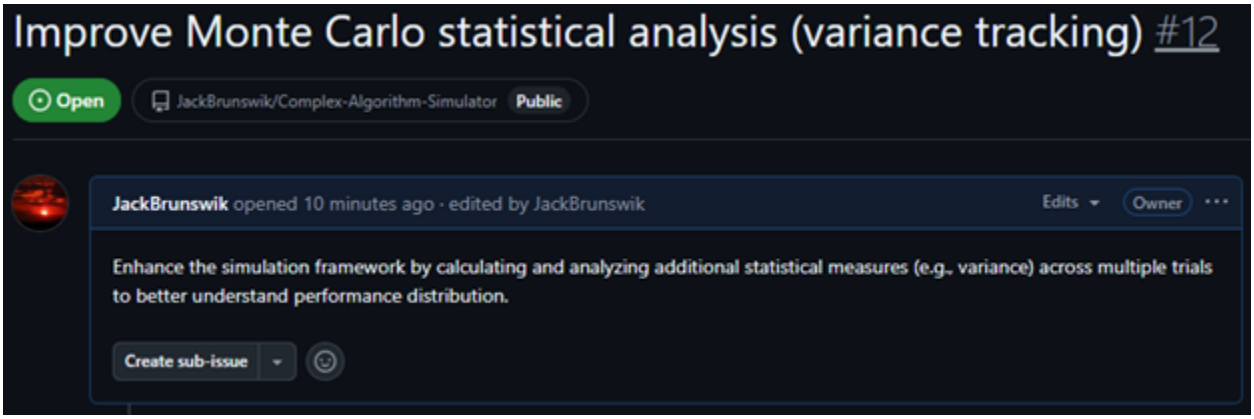

Figure 3: Brief description of in-progress work

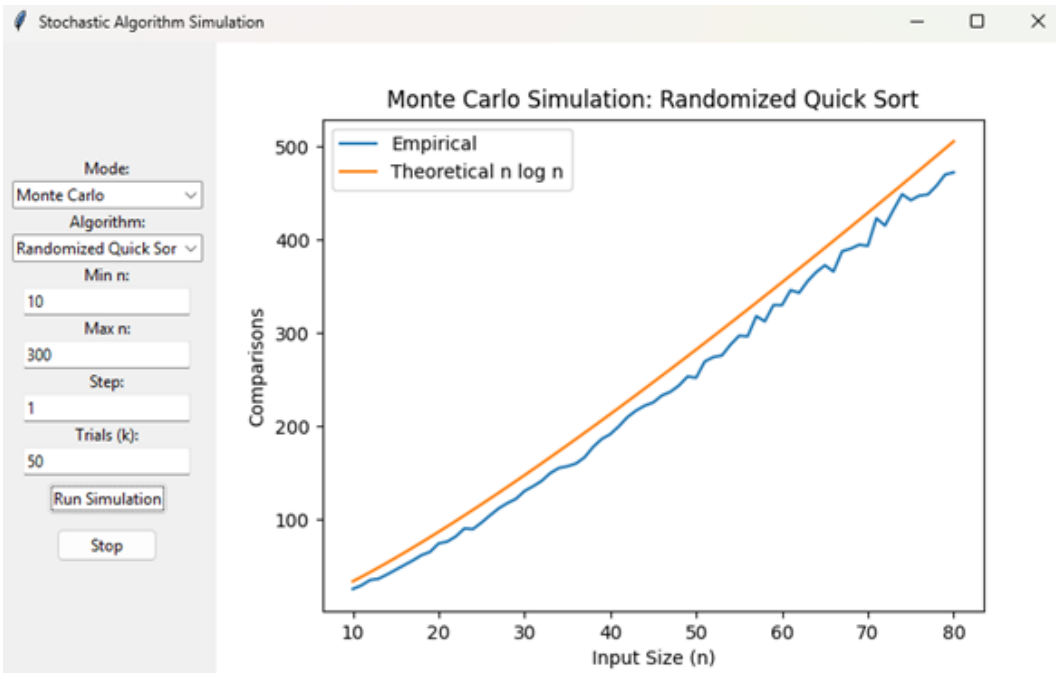

Figure 4: Randomized Quick Sort running in Monte Carlo mode - simulation

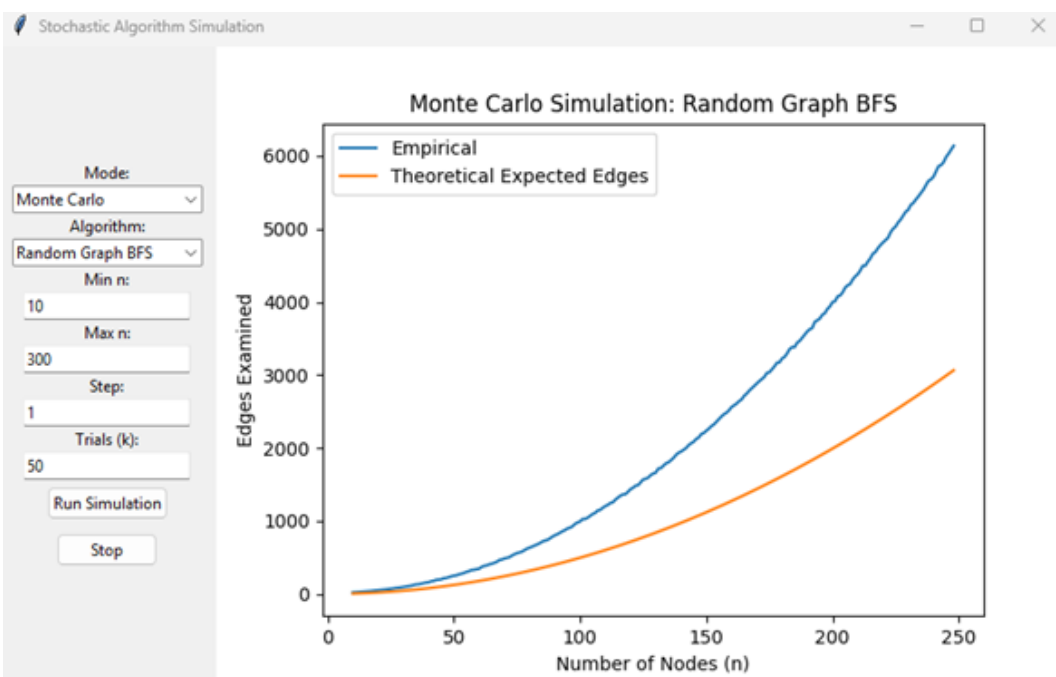

Figure 5: Random Graph BFS running in Monte Carlo mode - simulation

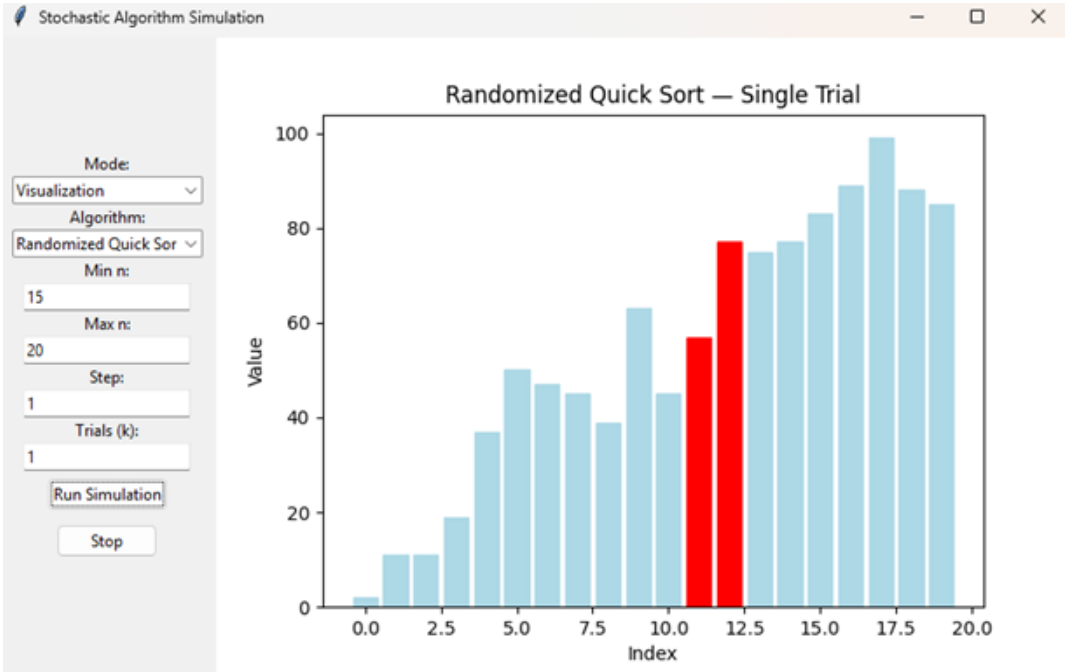

Figure 6: Randomized Quick Sort running in Visualization mode – single trial

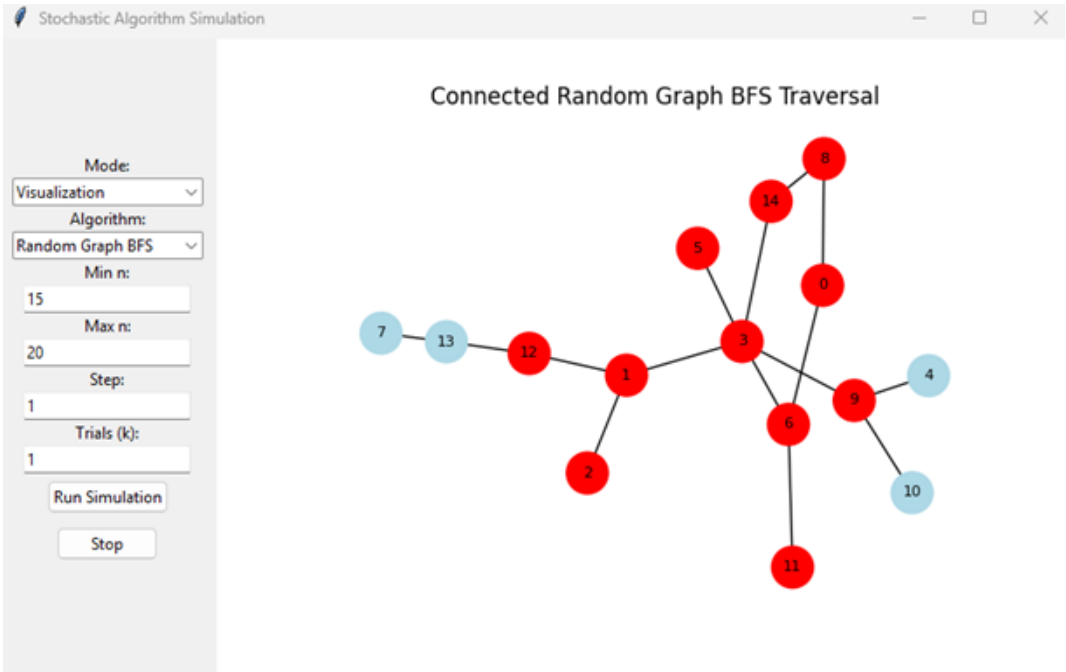

Figure 7: Random Graph BFS running in Visualization mode – single trial

Status Summary

The project is currently a functional stochastic algorithm simulation framework featuring two implemented models: Randomized QuickSort and Random Graph BFS using the Erdős–Rényi model. It supports both Monte Carlo experimentation and real-time visualization modes, allowing users to observe empirical performance growth and animated algorithm execution. Since the original proposal, the architecture was refined from a generalized event-driven design into a focused Monte Carlo-based simulator, with integrated metric collection and dynamic plotting using Matplotlib and NetworkX.

Key accomplishments since M1 include implementing structured performance tracking, real-time graphical updates, and dual-mode execution (statistical analysis and visualization). Challenges primarily involved simplifying the original UML architecture to better fit the project's scope and maintaining GUI responsiveness during simulations; these were resolved by merging responsibilities within the main controller and optimizing plotting logic. Next steps include enhancing statistical analysis (variance tracking and histogram mode) and evaluating the integration of additional randomized algorithms such as Karger's min-cut to deepen the simulation capabilities.